

RESEARCH METHODOLOGY

PREPROCESSING AND ANALYSIS OF DATA

Presented by **DR TUMPA BANERJEE**



www.sittechno.org

22-09-2026

Dr. Tumpa Banerjee



INTRODUCTION TO DATA PROCESSING & ANALYSIS

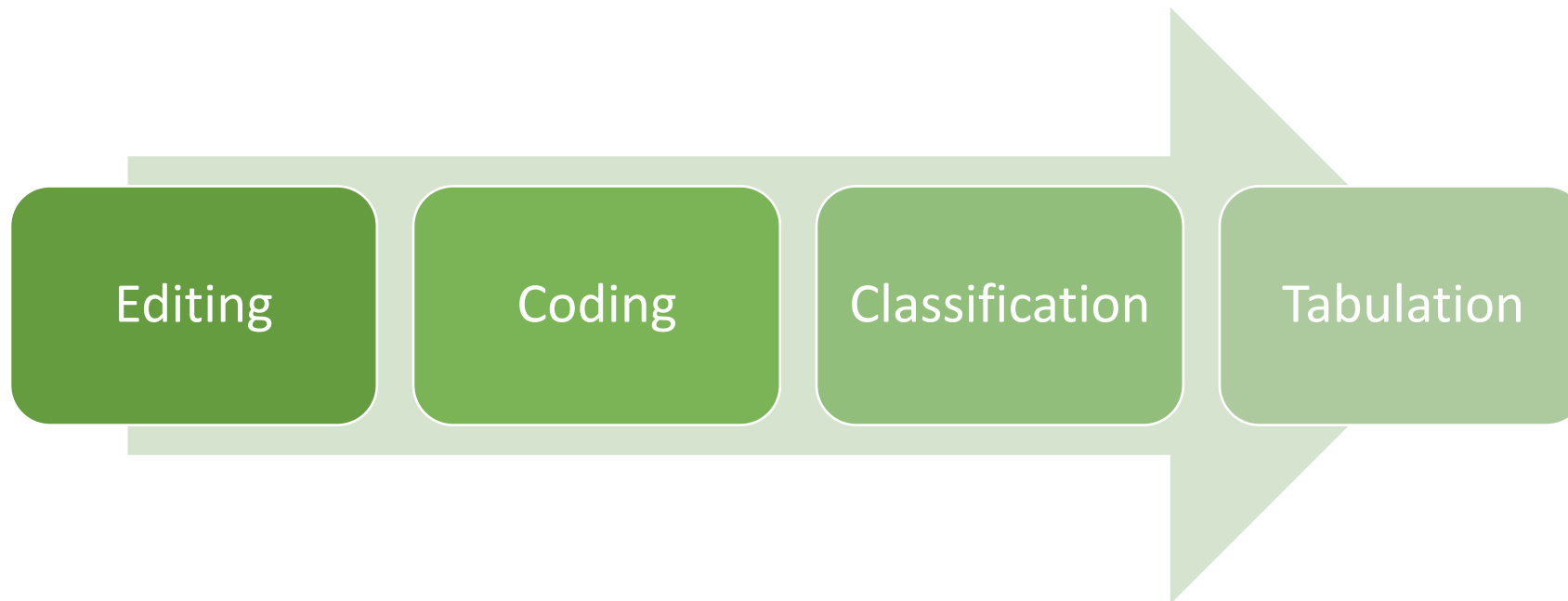
Raw data collected from the field requires strategic manipulation to become meaningful statement summaries.

The term "data processing" describes a number of procedures, including coding, editing, scoring computation, creating master charts, etc. Each and every step of the research process requires a strategy from the researcher.

- **Data Processing:** The conversion of raw data into a structured format suitable for analysis.
- **Data Analysis:** Computation of certain measures, patterns, and relationships among data groups to test hypotheses and draw conclusions.

THE FOUR CORE OPERATIONS OF DATA PREPROCESSING

4 interconnected blocks or a sequential flowchart:



These four operations clean and arrange raw facts so they are ready for definitive statistical testing.

DATA EDITING: CLEANING THE RAW METRICS

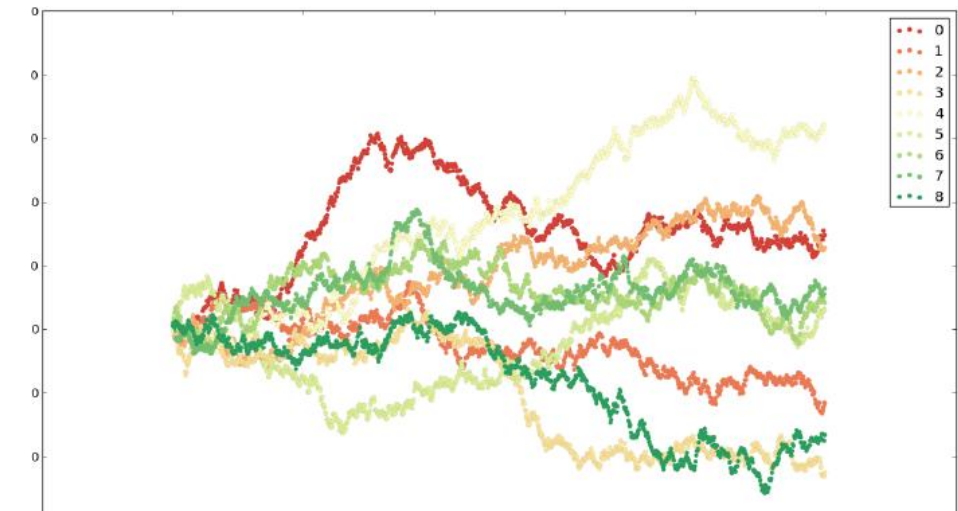
- **Definition:** Examining collected raw data to detect and correct errors, omissions, and inconsistencies.
- **Field Editing:** Reviewing reporting forms immediately after collection by the investigator to fix illegible or incomplete responses.
- **Central Editing:** Comprehensive in-house scrubbing carried out when all forms are returned to the central office.

DATA CODING: ASSIGNING STRUCTURED VALUES

- **Definition:** Assigning numerals or other symbols to answers so responses can be put into a limited number of categories.
- **Categorization:** Reduces massive text variations into uniform classes appropriate for efficient tabulation.
- **Pre-Coding:** Incorporating code symbols directly into the questionnaire template before it is sent out to respondents.

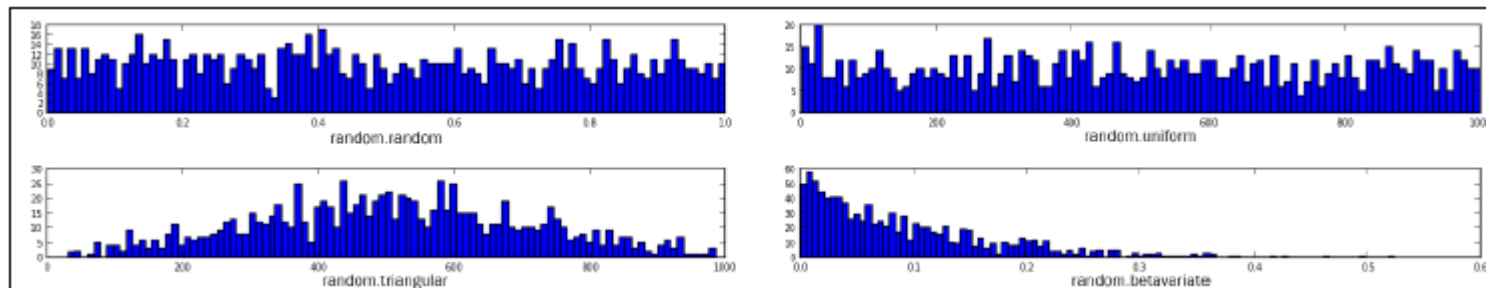
CLASSIFICATION: GROUPING THE INFORMATION

- **Definition:** Arranging data into groups or classes based on shared characteristics.
- **Classification by Attributes:** Categorizing based on descriptive qualities (e.g., literacy, gender, region) which cannot be measured quantitatively.
- **Classification by Class Intervals:** Categorizing based on numerical quantities (e.g., income ranges, age groups, weight brackets)



TABULATION: SUMMARIZING TO TABLES

- **Definition:** Orderly arrangement of data in rows and columns to present a concise overview of findings.
- **Purpose:** Conserves space, simplifies structural comparisons, and makes calculation omissions easier to detect.
- **Essential Components:** Every table must feature a clear Title, precise Row Headings (Stubs), Column Headings (Captions), and data Sources.



TRANSFORMING PROCESSED DATA INTO INSIGHTS

- **Objective:** To determine whether the collected data confirms or refutes the initial hypothesis.
- **Descriptive Analysis:** Shifting data down to a single index to summarize its basic features (frequencies, percentages).
- **Inferential Analysis:** Using sample statistics to make judgments and estimate the characteristics of an entire population.

KEY STATISTICAL MEASURES

Category	Analytical Method/measurement	Purpose
Central Tendency	Mean, Median, Mode	Locates the center or typical value of a distribution.
Dispersion	Range, Variance, Standard Deviation	Measures how scattered or spread out data values are.
Assymetry	Skewness and Kurtosis	Determines if data distribution is uniform or shifted.
Relationship	Correlation and Regression	Evaluates dependencies and links between variables.

TESTING OF HYPOTHESES

- **Parametric Tests:** Applied to normal distributions; includes **Z-tests, t-tests, and F-tests (ANOVA)**.
- **Non-Parametric Tests:** Distribution-free tests applied to nominal or ordinal data; includes the **Chi-Square (χ^2) test**.
- **Decision Matrix:** If the calculated test value exceeds the critical value at a chosen level of significance, the null hypothesis (H_0) is rejected

$$\text{HSD} = q\sqrt{\frac{\text{MSE}}{n}} = 5.04\sqrt{\frac{1.63}{5}} = 2.88$$

CONCLUSION

- **Garbage In, Garbage Out:** Bad data preprocessing leads to flawed statistical conclusions.
- **Systematic Progression:** Moving seamlessly from field editing to advanced hypothesis tests ensures research integrity.
- **Final Output:** Clean, well-analyzed data creates a smooth transition to interpretation and formal report writing.

THANK YOU

I APPRECIATE YOUR TIME AND
ATTENTION TO OUR RESEARCH STUDY.

Presented by **DR TUMPA BNAERJEE**

📞 +91 9883057767

🌐 www.sittechno.edu.in

✉️ Tmp.manna@gmail.com

